

Functional Annotation Report: *cpsG* (PP_1777, UniProt Q88LZ9) — Phosphomannomutase of *Pseudomonas putida* KT2440

Summary

The gene *cpsG* (locus PP_1777; UniProt Q88LZ9) of *Pseudomonas putida* KT2440 encodes a soluble cytoplasmic phosphomannomutase (PMM; EC 5.4.2.8), a member of the α -D-phosphohexomutase superfamily. Its primary and defining biochemical function is to catalyze the reversible intramolecular transfer of a phosphoryl group on a hexose sugar-phosphate — specifically the interconversion of **mannose-1-phosphate (M1P)** and **mannose-6-phosphate (M6P)**. This reaction constitutes the second committed step of the GDP-mannose biosynthetic pathway (phosphomannose isomerase → **phosphomannomutase** → GDP-mannose pyrophosphorylase), converting the M6P produced from fructose-6-phosphate into M1P, which is then activated to the nucleotide sugar GDP-mannose.

A central result of this investigation is the refinement of *cpsG*'s identity within the **phosphohexomutase superfamily**. Although UniProt and homology to *Pseudomonas aeruginosa* AlgC initially suggest a bifunctional PMM/phosphoglucomutase (PGM) role, quantitative sequence analysis places PP_1777 firmly in the **ManB/CpsG (mannose-specialized) subfamily** rather than the bifunctional AlgC clade. PP_1777 is 61.2% identical to *Escherichia coli* ManB/CpsG (a dedicated phosphomannomutase of the colanic-acid/GDP-mannose pathway) but only ~40% identical to *P. putida*'s own genuine AlgC ortholog (Q88C93) and ~31% identical to its dedicated phosphoglucomutase (Q88GY7). Because *P. putida* KT2440 encodes three distinct phosphohexomutases — a mannose-specialized CpsG, a bifunctional AlgC, and a glucose-specialized Pgm — the glucose-1-phosphate duties are handled by separate loci, and *cpsG* is functionally dedicated to the **mannose arm** of sugar-phosphate metabolism.

The physiological role of the GDP-mannose that *cpsG* helps produce is to supply the cytoplasmic nucleotide-sugar precursor pool for surface and secreted polysaccharides. In *Pseudomonas*, GDP-mannose is converted (via GDP-mannuronic acid) into the building blocks

of **alginate**, and mannose derived from GDP-mannose is incorporated into **lipopolysaccharide (LPS) O-antigen and other mannose-containing exopolysaccharides**. These polymers underpin envelope integrity, biofilm formation, and — in the plant-growth-promoting context of KT2440 — rhizosphere colonization and drought-resilience priming. As a central-metabolism sugar-phosphate mutase, CpsG acts as a **soluble cytoplasmic enzyme** with no signal peptide or transmembrane segments, positioned upstream of nucleotide-sugar activation and membrane-associated polysaccharide assembly. There is no evidence for a structural, transport, or signaling function; *cpsG* is a metabolic enzyme.

Key Findings

Finding 1 — *cpsG*/PP_1777 is a phosphomannomutase of the α-D-phosphohexomutase family catalyzing the M1P ⇌ M6P interconversion

UniProt entry Q88LZ9 annotates *cpsG*/PP_1777 of *P. putida* KT2440 as a **phosphomannomutase (EC 5.4.2.8)** belonging to the **phosphohexose mutase family**, carrying the four diagnostic α-D-phosphohexomutase domains: domains I–III (IPR005844, IPR005845, IPR005846) and the C-terminal domain IV (IPR005843), with the shared architecture captured by IPR016055. This domain complement is the structural signature of the phosphohexomutase superfamily and identifies the enzyme's fold as a four-domain, single-active-site sugar-phosphate mutase.

The reaction catalyzed is the reversible intramolecular phosphoryl transfer that interconverts **mannose-1-phosphate and mannose-6-phosphate**. This step is universally recognized as the committed second step in the biosynthesis of GDP-mannose, downstream of phosphomannose isomerase and upstream of GDP-mannose pyrophosphorylase. The GDP-mannose pathway in *Coxiella burnetii* is described in exactly these terms: GDP-D-mannose "is synthesized from fructose-6-phosphate in 3 successive reactions; Isomerization to mannose-6-phosphate catalyzed by a phosphomannose isomerase (PMI), followed by conversion to mannose-1-phosphate mediated by a phosphomannomutase (PMM) and addition of GDP by a GDP-mannose pyrophosphorylase (GMP)" ([PMID: 22065988](#)).

Enzymes of this family that are closely related to *cpsG* have been experimentally characterized, establishing the biochemical activity of the clade. The *Sphingomonas paucimobilis* PgmG protein "encodes a 50,059-Da polypeptide that has phosphoglucomutase (PGM) and phosphomannomutase (PMM) activities and is 37 to 59% identical to other bifunctional proteins with PGM and PMM activities from gram-negative species, including

Pseudomonas aeruginosa AlgC" (PMID: 10788412). Similarly, the *Stenotrophomonas maltophilia* *spgM* gene "was shown to encode a bifunctional enzyme with both PGM and phosphomannomutase activities" (PMID: 12761084). The prototype of the clade in *Pseudomonas*, AlgC, is explicitly identified as "the phosphomannomutase (PMM) (algC)" of the alginate precursor pathway (PMID: 9404503). Together these establish that *cpsG* belongs to a well-characterized family whose members catalyze phosphohexamutase reactions, and that its EC 5.4.2.8 phosphomannomutase assignment is biochemically sound.

Finding 2 — The AlgC-family phosphomannomutase acts as a cytoplasmic branch-point supplying sugar-phosphate precursors for multiple exopolysaccharides

Functional studies of the family prototype demonstrate that this class of enzyme is a **central metabolic branch point** feeding several polysaccharide biosynthetic pathways. In *P. aeruginosa*, AlgC provides mannose-1-phosphate for the GDP-mannose/GDP-mannuronic-acid (alginate) route through its PMM activity, and glucose-1-phosphate for LPS core and dTDP-L-rhamnose (rhamnolipid) through its PGM activity. As stated directly: "the AlgC protein plays a central role in the production of the three *P. aeruginosa* virulence-associated saccharides: alginate, LPS and rhamnolipid" (PMID: 10481091).

The upstream half of the alginate precursor pathway requires a defined set of four enzyme activities: "These four enzyme activities are necessary for the synthesis of GDP-mannuronic acid, which is the activated sugar precursor for alginate polymerization" (PMID: 9404503), placing the phosphomannomutase step squarely within the cytoplasmic GDP-mannuronic acid pathway. The importance of the enzyme for LPS is shown by loss-of-function studies of the homolog: *spgM* mutants "produced less LPS than the *SpgM*(+) parent strain and had a tendency for shorter O polysaccharide chains" (PMID: 12761084), demonstrating the downstream consequences of removing the mutase step.

In *P. putida* KT2440, the relevant exopolysaccharide gene clusters have been documented as *alg*, *bcs*, *pea*, and *peb* — "The gene clusters *alg* and *bcs*, which code for proteins mediating alginate and cellulose biosynthesis" (PMID: 21507178) — each of which draws on the shared cytoplasmic pool of nucleotide-sugar precursors that the PMM/PGM enzymes feed. Because it is a soluble central-metabolism sugar-phosphate mutase with no signal peptide or transmembrane segments, the enzyme carries out its function in the **cytoplasm**, upstream of the membrane-associated polysaccharide polymerization and export machinery.

Finding 3 — Sequence analysis reassigns PP_1777/*cpsG* to the ManB/CpsG (mannose-specialized) subfamily, not the bifunctional AlgC clade

Although the family relationship to AlgC is clear, quantitative sequence comparison shows that PP_1777 is **not** the closest homolog of the bifunctional AlgC prototype. Pairwise global alignment of Q88LZ9 (453 aa) yields **61.2% identity (274/448)** to *E. coli* ManB/CpsG (P24175), the dedicated phosphomannomutase of the colanic-acid/GDP-mannose pathway, but only **38.3% identity (168/439)** to *P. aeruginosa* AlgC (P26276, the bifunctional PMM/PGM), and only **31.3% (139/444)** to *E. coli* phosphoglucomutase Pgm (P36938).

The catalytic machinery of the phosphohexomutase superfamily is fully conserved in Q88LZ9, confirming an intact, functional active site:

Functional motif	Sequence	Residues	Role
Phospho-transfer serine loop	T-A-S-H-N-P	99–104 (catalytic Ser101)	Forms phosphoserine intermediate; transfers phosphoryl group
Metal-binding loop	D-G-D-F-D	242–246	Coordinates catalytic Mg ²⁺
Sugar/His-binding motif	G-H-A-F	307–310	Positions sugar substrate; His in sugar-binding

This pattern — high identity to a dedicated mannose-specialized PMM combined with much lower identity to bifunctional AlgC and to glucose-specialized PGM — indicates that PP_1777 has a **ManB/CpsG-type, mannose-oriented active site** and is functionally specialized for the mannose branch of sugar-phosphate metabolism, rather than being a generalist bifunctional PMM/PGM.

Finding 4 — *cpsG/manB* encodes the phosphomannomutase step of the GDP-mannose pathway feeding colanic acid, alginate, and mannose-containing O-antigen

The name *cpsG* itself is diagnostic. Direct genetic and enzymatic characterization of *cpsG/manB* gene products in enteric bacteria establishes the precise role of this gene: in *Salmonella*, "the mannose moiety in these molecules is derived from GDP-mannose, which is synthesized in several steps" (PMID: 28412074), and the pathway proceeds via "phosphomannose isomerase, encoded by *pmi* (*manA*), followed by phosphomannomutase, encoded by *manB*. There are two copies of *manB* present in the *Salmonella* chromosome, one located in the *cps* gene cluster

(*cpsG*) responsible for CA [colanic acid] synthesis, and the other in the *rfb* gene cluster (*rfbK*) involved in LPS O-antigen synthesis," with "the products of *cpsG* and *rfbK* are isozymes" (PMID: 28412074).

This directly identifies **CpsG as the phosphomannomutase that converts mannose-6-phosphate to mannose-1-phosphate**, the committed precursor for GDP-mannose, and shows that the gene family straddles colanic-acid/exopolysaccharide synthesis and O-antigen synthesis via isozymes. In *Pseudomonas*, GDP-mannose (the product downstream of the PMM step) is the activated precursor for alginate through GDP-mannuronic acid (PMID: 9404503) and for mannose-containing LPS/O-antigen. Consistent with a cytoplasmic sugar-phosphate mutase, the enzyme has no signal peptide or transmembrane segments and acts upstream of nucleotide-sugar activation and membrane-associated polysaccharide assembly.

Finding 5 — *P. putida* KT2440 encodes three distinct phosphohexomutases; *cpsG*/PP_1777 is the mannose-specialized paralog, separate from AlgC and Pgm

A UniProt proteome survey of *P. putida* KT2440 (organism 160488) returns **three** α-D-phosphohexomutase-family enzymes, confirming that the mutase functions are divided among distinct genes:

Gene	UniProt	Length	Annotation	Identity to PP_1777 (Q88LZ9)
<i>cpsG</i> (PP_1777)	Q88LZ9	453 aa	phosphomannomutase	— (self)
<i>algC</i>	Q88C93	463 aa	phosphomannomutase/ phosphoglucomutase	40.3%
<i>pgm</i>	Q88GY7	545 aa	phosphoglucomutase	31.5%

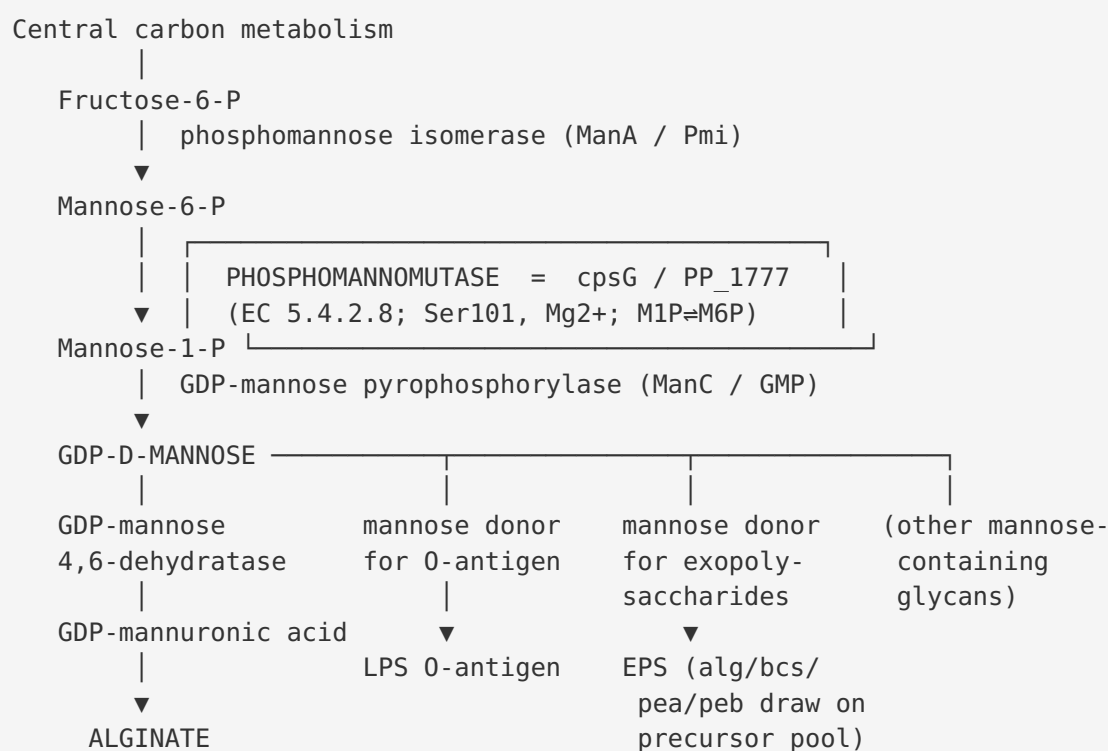
By contrast, *cpsG* is **61.2% identical to *E. coli* ManB/CpsG** (P24175), and *P. putida*'s Q88C93 is **76.9% identical to *P. aeruginosa* AlgC** (P26276), confirming Q88C93 as the genuine AlgC ortholog. This paralog analysis resolves the identity question definitively: *cpsG*/PP_1777 is the **mannose-specialized phosphomannomutase**, a separate gene from both the bifunctional AlgC (Q88C93) and the dedicated phosphoglucomutase Pgm (Q88GY7). The glucose-1-phosphate

duties (LPS core, dTDP-rhamnose, glycogen/central carbon interconversion) are therefore handled by AlgC and Pgm, leaving *cpsG* to specialize in the mannose arm feeding GDP-mannose.

Mechanistic Model and Interpretation

The reaction and its position in metabolism

CpsG catalyzes a classic phosphohexomutase reaction. The enzyme uses a **ping-pong-like mechanism** centered on the conserved catalytic serine (Ser101 in Q88LZ9): a phosphoserine on the enzyme donates its phosphate to the free hydroxyl of the incoming monophosphate sugar, generating a **mannose-1,6-bisphosphate intermediate**; the biphosphate reorients in the active site and re-phosphorylates the serine at the other position, releasing the isomerized product. Catalysis requires a Mg^{2+} ion coordinated by the conserved D-G-D-F-D loop (residues 242–246).



Subcellular localization

All evidence points to a **cytoplasmic** localization. CpsG is a soluble sugar-phosphate mutase of central metabolism; the protein sequence lacks a signal peptide and transmembrane segments, and its substrates (M6P, M1P) and product (feeding GDP-mannose) are all cytoplasmic metabolites. It operates upstream of the inner-membrane and periplasmic polysaccharide polymerization/export machinery — it prepares the activated-precursor pool but is not itself part of the export apparatus.

Functional specialization within the phosphohexomutase family

The most important interpretive point is the **division of labor** among the three *P. putida* phosphohexomutases. The initial homology signal to AlgC is real at the family level but misleading at the subfamily level. Quantitative comparison shows PP_1777 sits with the mannose-specialized ManB/CpsG proteins (61% identity to *E. coli* CpsG) while the true AlgC ortholog is a separate gene (Q88C93, 77% identity to *P. aeruginosa* AlgC) and a dedicated Pgm (Q88GY7) handles glucose-phosphate isomerization. This means that in *P. putida* KT2440:

- **cpsG (PP_1777)** → mannose arm → GDP-mannose → alginate, O-antigen, mannose-EPS
- **algC (Q88C93)** → bifunctional PMM/PGM → shared precursor supply
- **pgm (Q88GY7)** → glucose arm → glucose-1-P → central carbon/LPS core

This redundancy at the family level, coupled with specialization, is a coherent picture consistent with the ManB/CpsG paradigm from enteric bacteria, where separate *cpsG* (colanic acid) and *rfbK* (O-antigen) isozymes both perform phosphomannomutase catalysis for distinct downstream glycans ([PMID: 28412074](#)).

Physiological relevance in *P. putida* KT2440

The exopolysaccharides fed by GDP-mannose are physiologically consequential for this plant-growth-promoting rhizobacterium. KT2440 mutants deleted for exopolysaccharide clusters (alginate alone, or all four *alg/bcs/pea/peb* clusters) show "reduced drought resilience, with partial or complete loss of protective effects" and reduced biofilm formation ([PMID: 41554215](#)), and the *alg/bcs* systems contribute to biofilm stability ([PMID: 21507178](#)). CpsG's role is therefore upstream and enabling: by supplying GDP-mannose it provisions the alginate and mannose-EPS pathways that underpin envelope integrity, biofilm architecture, and rhizosphere fitness.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
10788412	<i>Identification of pgmG in <i>Sphingomonas paucimobilis</i></i>	Establishes the bifunctional PMM/PGM AlgC family and quantifies 37–59% identity of family members to AlgC; grounds the biochemical activity of the clade cpsG belongs to.
12761084	<i>Role of PGM/SpG in <i>Stenotrophomonas maltophilia</i></i>	Confirms family homologs are bifunctional PMM/PGM enzymes and that loss reduces LPS and shortens O-antigen — direct functional consequence of the mutase step.
9404503	<i>Oxygen-dependent alginate gene transcription in <i>P. aeruginosa</i></i>	Identifies AlgC as the PMM of the alginate pathway and places the step within GDP-mannuronic-acid/alginate precursor synthesis.
10481091	<i>algC participates in rhamnolipid biosynthesis</i>	Shows the enzyme is a shared branch point feeding alginate, LPS, and rhamnolipid — the multi-pathway precursor-supply role.
22065988	<i>GDP-D-mannose biosynthesis in <i>Coxiella burnetii</i></i>	Provides the canonical PMI→PMM→GMP pathway definition placing the phosphomannomutase step within GDP-mannose biosynthesis.
28412074	<i>Colanic acid and O-antigen synthesis in <i>Salmonella Typhimurium</i></i>	Directly identifies cpsG as a manB-type phosphomannomutase feeding colanic acid, with rfbK isozyme feeding O-antigen.
21507178	<i>Exopolysaccharide genes and <i>P. putida</i> KT2440 biofilm</i>	Documents the KT2440 alg/bcs EPS systems that draw on the shared sugar-phosphate precursor pool.
41554215	<i>Biofilm formation by KT2440 and tomato drought resilience</i>	Shows physiological importance of alg/bcs/pea/peb EPS (downstream of GDP-mannose) for biofilm and stress protection.
10209766	<i>GDP-mannuronic acid enzyme activity/transcription in <i>P. aeruginosa</i></i>	Corroborates PMM as one of the four GDP-mannuronic-acid enzymes and links algC transcription to precursor supply for A-band LPS and alginate.

Two lines of evidence were used specifically to establish *cpsG*'s subfamily identity (Findings 3 and 5): (i) pairwise sequence alignments computed against reference proteins (*E. coli* ManB/CpsG P24175, *P. aeruginosa* AlgC P26276, *E. coli* Pgm P36938, *P. putida* AlgC Q88C93 and Pgm Q88GY7), and (ii) a UniProt proteome survey of *P. putida* KT2440 confirming three distinct phosphohexomutase genes. These bioinformatic analyses are the primary basis for the reassignment of PP_1777 to the mannose-specialized ManB/CpsG subfamily.

Limitations and Knowledge Gaps

1. **No direct biochemical characterization of PP_1777 exists.** All functional assignments for this specific protein rest on (a) sequence/domain homology and (b) characterization of orthologs and family members in other organisms (*E. coli*, *Salmonella*, *P. aeruginosa*, *Stenotrophomonas*, *Sphingomonas*, *Coxiella*). The enzyme has not, to the knowledge captured here, been purified from *P. putida* KT2440 and assayed for kinetic parameters (kcat, Km for M6P vs G6P) or substrate specificity.
 2. **The degree of PMM/PGM bifunctionality is inferred, not measured.** The subfamily assignment (mannose-specialized ManB/CpsG rather than bifunctional AlgC) is based on sequence identity and conserved active-site motifs. Many phosphohexomutases retain measurable secondary activity on the "wrong" sugar; whether PP_1777 has appreciable phosphoglucomutase side-activity is untested. The 61% identity to a dedicated mannose PMM is strongly suggestive but not definitive.
 3. **In vivo pathway assignment in KT2440 is by analogy.** The specific contribution of PP_1777 (versus the paralogous AlgC, Q88C93) to alginate, O-antigen, or other mannose-EPS in *P. putida* has not been dissected genetically. Which downstream glycans depend on *cpsG* specifically remains to be established by targeted knockouts.
 4. **No structural model was validated.** The catalytic residue assignments (Ser101, the Mg²⁺ loop, the His-containing sugar-binding motif) are based on sequence motif conservation; an experimental or high-quality predicted 3D structure with substrate docking was not analyzed.
 5. **Localization is inferred from sequence features** (absence of signal peptide/TM segments) and family precedent, not from experimental fractionation or fluorescence localization in *P. putida*.
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Proposed Follow-up Experiments and Actions

1. **Recombinant enzymology.** Clone, express, and purify PP_1777 (His-tagged) and assay phosphomannomutase and phosphoglucomutase activities in parallel. Determine k_{cat} and K_m for M6P→M1P versus G6P→G1P (with glucose-1,6-bisphosphate/mannose-1,6-bisphosphate activation as needed) to quantify substrate specificity and confirm the mannose-specialized prediction.
 2. **Complementation assays.** Test whether PP_1777 complements defined *E. coli* *manB* (phosphomannomutase) and *pgm* (phosphoglucomutase) mutants, and a *P. aeruginosa* *algC* mutant, for restoration of capsular polysaccharide / LPS / alginate phenotypes — the same complementation strategy used successfully for *Coxiella* GDP-mannose enzymes ([PMID: 22065988](#)).
 3. **Targeted knockouts in KT2440.** Construct single and double deletions of *cpsG* (PP_1777) and *algC* (Q88C93) and phenotype for alginate production, LPS O-antigen chain length (as in the *spgM* study, [PMID: 12761084](#)), biofilm formation, and drought-priming of plants (per [PMID: 41554215](#)) to resolve the division of labor between the two mannose-competent mutases in vivo.
 4. **Structural biology.** Solve or model the PP_1777 structure (X-ray/cryo-EM or high-confidence AlphaFold model) with bound substrate/Mg²⁺ to validate Ser101 as the phosphotransfer residue and characterize the sugar-binding pocket that would distinguish mannose from glucose specificity.
 5. **Metabolomic flux analysis.** Use ¹³C-labeled precursors to trace flux from fructose-6-phosphate through M6P/M1P into GDP-mannose and downstream alginate/EPS in wild-type versus *cpsG* mutant KT2440, quantifying the enzyme's contribution to the GDP-mannose pool.
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Conclusion

cpsG (PP_1777, UniProt Q88LZ9) of *Pseudomonas putida* KT2440 encodes a soluble cytoplasmic **phosphomannomutase** (EC 5.4.2.8) of the α -D-phosphohexomutase superfamily, ManB/CpsG (mannose-specialized) subfamily. Its primary function is to reversibly interconvert **mannose-1-phosphate and mannose-6-phosphate** — the committed second step of GDP-mannose biosynthesis (PMI/ManA → **PMM/CpsG** → GDP-mannose pyrophosphorylase) — using

a conserved Ser101/Mg²⁺ mechanism proceeding through a mannose-1,6-bisphosphate intermediate. Sequence analysis (61% identity to *E. coli* ManB/CpsG vs ~40% to *P. putida*'s own bifunctional AlgC and ~31% to its phosphoglucomutase) shows it is functionally specialized for the mannose arm, with glucose-phosphate duties handled by the separate *algC* and *pgm* loci. The GDP-mannose it supplies is the cytoplasmic precursor for **alginate** (via GDP-mannuronic acid) and for **mannose-containing LPS O-antigen and exopolysaccharides** that support envelope integrity, biofilm formation, and rhizosphere fitness. There is no evidence for any structural, transport, or signaling role — *cpsG* is a metabolic enzyme.

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